Charge removal opens product-forbidden rank channels
An exact correction to the memory-filter ansatz

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Abstract

The collinearity of four quadratic maps in the previously tested $N = 5, p = 2, m = 3$ system does not extend to general charge sectors. At half filling, particle-hole transposition imposes an exact parity selection rule on symmetrized products. Charge removal can populate the forbidden channels. We give an analytic family for the force norm A and an independently verified counterexample for the actual quadratic contraction of the quartic memory core. A two-charge amplitude representation and an exact norm identity for charge removal supply the corrected starting point for the spectral limit.

1 A selection rule for the multiplication map

Use standard CAR and the full overlapping complex Gaussian p-body Hamiltonian, $\mathbb{E}J_{IJ}J_{KL} = w\delta_{IL}\delta_{JK}$, with $\hbar = 1$. On charge m, tr_m denotes normalized trace and P_r the neutral $U(N)$ rank projector. All statements through Section 4 are finite-dimensional identities, with no double-scaling limit.

For $N = 2m$, let $\star$ be the signed complement map on $\wedge^m\mathbb{C}^N$, and define the linear anti-automorphism

$$\tau(X) = \star X^{\mathsf{T}} \star^{-1}, \qquad \tau(XY) = \tau(Y)\tau(X), \qquad \tau^2 = 1. \tag{1}$$

It sends $E_{ij} = c_i^\dagger c_j$ to $\delta_{ij} - E_{ij}$. Consequently it commutes with the adjoint $U(N)$ action. The neutral operator representation is multiplicity-free in rank. On a disjoint, traceless rank-r highest-weight monomial, reversing and normal ordering gives $(-1)^r$. Thus

$$\tau P_r = (-1)^r P_r. \tag{2}$$

For arbitrary pure-rank X_a, Y_b, it follows that

$$P_r\{X_a, Y_b\} = 0 \quad \text{if } a + b - r \text{ is odd.} \tag{3}$$

This is a statement about the multiplication map, not the absence of the irrep from the tensor product. In particular, every odd-rank component of H_a^2 vanishes. An equivariant quadratic map need not satisfy the same selection rule.

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2 A family of pure-rank counterexamples

Let $n_i = c_i^\dagger c_i$, $1 \le p < m = N/2$, and choose the deterministic p-body input

$$H = \prod_{a=1}^{p} d_a, \qquad d_a = n_{2a-1} - n_{2a}, \quad t_a = n_{2a-1} + n_{2a}, \quad b_a = d_a^2. \tag{4}$$

Every partial contraction of its p-body coefficient tensor vanishes: the two choices in the omitted pair cancel. Hence this input is pure rank p wherever its embedding is nonzero. Also $\tau H = (-1)^p H$ and $H^2 = \prod_a b_a$ is even.

Direct removal of an occupied mode gives

$$A = \sum_i [H, c_i]^\dagger [H, c_i] = \sum_{a=1}^{p} t_a \prod_{b \ne a} b_b. \tag{5}$$

Under complementation, $t_a \mapsto 2 - t_a$ and $b_a \mapsto b_a$. Its odd part is therefore

$$A_{\text{odd}} := \frac{A - \tau A}{2} = \sum_{a=1}^{p} (t_a - 1) \prod_{b \ne a} b_b \ne 0. \tag{6}$$

A configuration with one double-occupied pair and all other selected pairs singly occupied has value $+1$; an empty exceptional pair gives -1. The $N - 2p$ remaining modes can accommodate the required charge because $m > p$. Every other configuration gives zero.

Counting these configurations yields exact normalized HS norms:

$$\|H^2\|_m^2 = \frac{2^p \binom{N-2p}{m-p}}{\binom{N}{m}}, \tag{7}$$

$$\|A_{\text{odd}}\|_m^2 = \frac{p\, 2^p \binom{N-2p}{m-p-1}}{\binom{N}{m}}, \tag{8}$$

$$\frac{\|A_{\text{odd}}\|_m^2}{\|H^2\|_m^2} = p\, \frac{m-p}{m-p+1}. \tag{9}$$

Thus no scalar coefficients multiplying the channels $P_r H_p^2$ can reproduce A for this family. It exists along half-filled double-scaled sequences as well, but these are deterministic inputs; Eq. (9) is not an annealed memory-strength calculation.

The checker exhausts all charge-m configurations in 15 systems, $N = 4, 6, 8, 10, 12$ and every $1 \le p < m$. It verifies the removal formula, both norm counts, and the rank eigenvalue $\mathcal{C}H = 2p(N + 1 - p)H$ using the diagonal Casimir action.

3 The actual quadratic memory contraction

The target is still the previously defined core

$$\mathcal{E} = \sum_i (\mathrm{ad}_H^2\, c_i)^\dagger\, \mathrm{ad}_H^2\, c_i + \tfrac{1}{4}[H,[H,A]], \qquad Q_{\mathcal{E}} = \tfrac{1}{2}\Delta\mathcal{E}. \tag{10}$$

For this homogeneous quartic polynomial, ordinary Gaussian chaos gives $W_2\mathcal{E} = Q_{\mathcal{E}} - 2\mathbb{E}\mathcal{E}$. The ensemble mean is scalar in each charge sector, so $P_1 W_2\mathcal{E} = P_1 Q_{\mathcal{E}}$.

Take $N = 6, p = 2, m = 3, w = 1$ and

$$h = (n_1 - n_2)(n_3 - n_4)\big|_3, \qquad k = (n_1 - n_2)(n_3 - n_4)\big|_2, \qquad D = n_1 + n_2 + n_3 + n_4 - 2I. \tag{11}$$

Both inputs have Casimir eigenvalue 20, hence rank two. Exact projection gives

$$\boxed{P_1 h^2 = 0, \qquad P_1 A = D, \qquad P_1 Q_{\mathcal{E}} = 99D.} \tag{12}$$

Here D has eigenvalues $+1$ on four states, -1 on four states, and zero on twelve states. In the normalized charge-three trace,

$$\|P_1 A\|_3^2 = \frac{2}{5}, \qquad \|P_1 Q_{\mathcal{E}}\|_3^2 = \frac{19602}{5}. \tag{13}$$

The old $J_h F$ direction cannot repair this zero within the $(2,2;1)$ block: on a pure input rank, the equivariant linear F is a scalar multiple of h, so $P_1 J_h F = 0$ as well.

The calculation uses the existing explicit $\Phi_\pm, \Psi, \mathrm{Rem}$ single-contraction formula. Independently, full quartic polarization of the commutator-defined core over all 225 real Gaussian coordinates reproduces $Q_{\mathcal{E}}$ exactly. The raw-word identity is checked too. Four full matrix Casimir projectors verify Eq. (2), not only its diagonal restriction.

This disproves the general scalar-filter ansatz. It does not disprove the previous $N = 5$ calculation or its exact Gram reconstruction. The present diagonal realization commutes with its projected D; it is not a new noncommuting dynamical witness. Neither a general multiplicity rank nor the double-scaled stochastic weight follows from this one realization.

4 The corrected two-charge interface

Write $h_a = H_m^{(a)}$, $k_a = H_{m-1}^{(a)}$ using the same coupling irrep. Define symmetric products as $X \odot Y = \{X, Y\}$ for distinct input ranks, and $X \odot X = X^2$ for equal ranks. Keep both maps

$$Z^+_{ab;r} = P_r(h_a \odot h_b), \qquad Z^-_{ab;r} = P_r \,\mathrm{Rem}(k_a \odot k_b). \tag{14}$$

Then the exact old identity $A = \mathrm{Rem}(k^2) + (2p - m)h^2$ becomes

$$A_{ab;r} = Z^-_{ab;r} + (2p - m)Z^+_{ab;r}. \tag{15}$$

No division by Z^+ is allowed. In reduced-tensor notation, choose primitive normal-ordered rank tensors $T_a^{(m)}$ with consistent normalization across charges, so $h_a = e_a(m)T_a^{(m)}$, $e_a(m) = \binom{m-a}{p-a}$. If $K^{(m)}_{ab;r,\rho}$ denotes their symmetrized product coefficient in a common intertwiner basis, the amplitude of A is

$$\begin{aligned} \mathsf{A}_{ab;r,\rho} &= (m - r)e_a(m - 1)e_b(m - 1)K^{(m-1)}_{ab;r,\rho} \\ &\quad + (2p - m)e_a(m)e_b(m)K^{(m)}_{ab;r,\rho}. \end{aligned} \tag{16}$$

This uses $\mathrm{Rem}(T_r^{(m-1)}) = (m - r)T_r^{(m)}$. The two reduced coefficients include their relative signs and multiplicity label ρ; the recoupling framework already exists in Kota–Vyas [1]. Their dilute limit is not the present limit.

There is also an exact norm transfer for every rank-r lower-sector operator X_r in the common rank range:

$$\|\,\mathrm{Rem}\, X_r\|_m^2 = \ell_r^2 \|X_r\|_{m-1}^2, \qquad \ell_r^2 = \frac{m(m - r)(N - m + 1 - r)}{N - m + 1} \le m^2. \tag{17}$$

To prove it, a disjoint rank-r monomial has norm proportional to $\binom{N-2r}{m-r}/\binom{N}{m}$. Taking the ratio between adjacent charges and multiplying by $(m - r)^2$ gives Eq. (17). Equivariance and irreducibility extend it to the full rank space. Eight direct CAR checks confirm it. Components outside the common rank range are killed by the equivariant removal map.

For $a < b$, the previous positive ordered-product law now gives

$$\mathbb{E}\|Z^-_{ab;r}\|_m^2 \le 4\ell_r^2 C_a^- C_b^- \pi_{ab}^{(m-1)}(r), \qquad C_a^- I = \mathbb{E}k_a^2. \tag{18}$$

For $a = b$ after W_2 centering, replace $4C_a^-C_b^-$ by $2(C_a^-)^2$. These are bounds; the same-coupling cross term $\mathbb{E}\langle Z^+, Z^-\rangle_m$ remains essential. The next calculation is therefore the coherent two-charge amplitudes of $Q_{\mathcal{E}}$, including Ψ insertions, before taking rank-deficit limits.

References

[1] V. K. B. Kota and M. Vyas, *Random Matrix Theory for Transition Strength Densities in Finite Quantum Systems*, Sec. 3, Eqs. (34)–(35), arXiv:1504.00759.

[2] V. K. B. Kota, *Bivariate moments of the two-point correlation function for embedded Gaussian unitary ensemble with k-body interactions*, Sec. IV, arXiv:2208.11312v2.